ECE 456 Problem Set 1:
The Basics of Electron Flow in Nanodevices

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1 Current with a Lorentzian Density of States

1.1 Alternative Form

Given the following expressions for the number of channel electrons and the current in a nanodevice:

$$N = \int_{-\infty}^{\infty} \frac{\gamma_1 f_1(E) + \gamma_2 f_2(E)}{\gamma_1 + \gamma_2} D(E-U)\ \mathrm{d}E \tag{1}$$

$$I = \frac{q}{\hbar} \frac{\gamma_1 \gamma_2}{\gamma_1 + \gamma_2} \int_{-\infty}^{\infty} [f_1(E) - f_2(E)] D(E-U)\ \mathrm{d}E \tag{2}$$

These expressions can be rewritten into a form that is more convenient for numerical implementation in MATLAB. First, define a change of variables:

$$\begin{aligned} E' &= E - U \\ E &= E' + U \\ \mathrm{d}E' &= \mathrm{d}E \end{aligned}$$

Since subtracting a constant does not alter the limits of integration, the integration bounds remain:

$$E' : -\infty \to \infty$$

Substituting $E = E' + U$ and relabeling $E' \to E$ yields:

$$N = \int_{-\infty}^{\infty} \frac{\gamma_1 f_1(E+U) + \gamma_2 f_2(E+U)}{\gamma_1 + \gamma_2} D(E)\ \mathrm{d}E \tag{3}$$

$$I = \frac{q}{\hbar} \frac{\gamma_1 \gamma_2}{\gamma_1 + \gamma_2} \int_{-\infty}^{\infty} [f_1(E+U) - f_2(E+U)] D(E)\ \mathrm{d}E \tag{4}$$

1.2 Solving the Equations Self-Consistently

Using the code provided in `problem1.py`, the values of N and I were computed self-consistently. The resulting plots are shown in Figure 1.

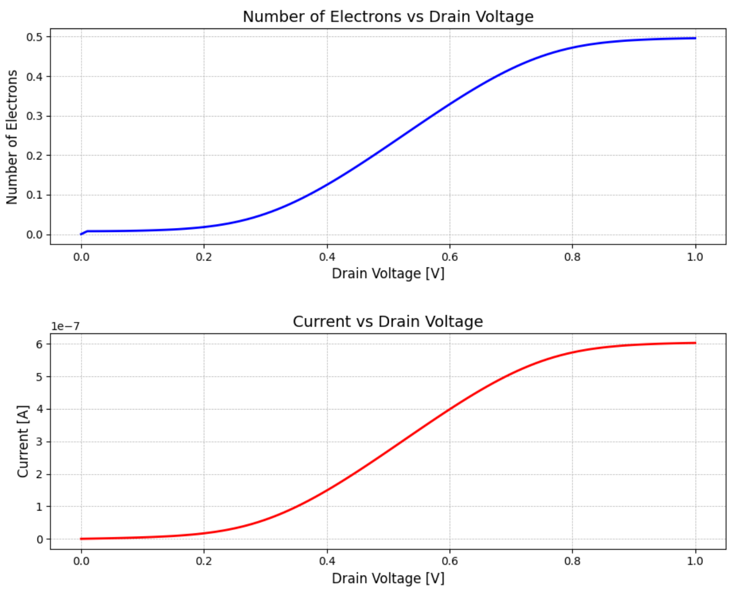


Figure 1: Current (I) and number of electrons (N) as functions of drain voltage (V_D).

1.3 Difference Proportional to Current

By plotting the difference between the Fermi functions using a discrete energy level (modeled as an impulse) rather than a broadened Lorentzian density of states, the relationship between the current I and the difference in the source and drain Fermi functions becomes clearer.

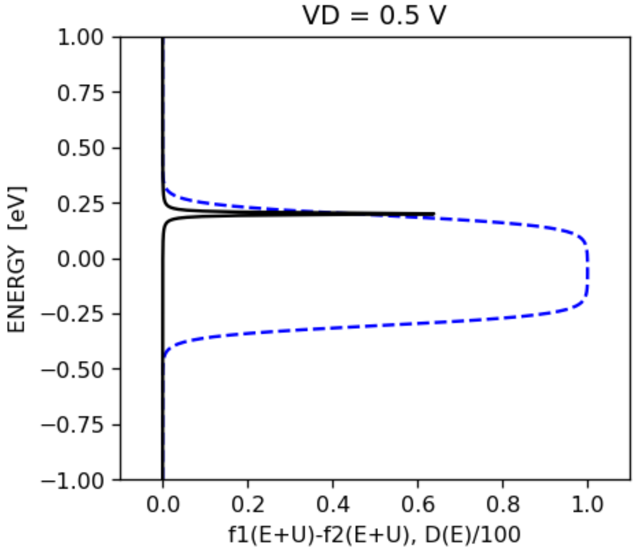


Figure 2: Difference between the Fermi functions as a function of energy, with the density of states modeled as an impulse.

V_D (V)	$[f_1(E+U) - f_2(E+U)]_{E=\mathcal{E}}$	I (A)
0.2	0.015	1.69×10^{-8}
0.5	0.440	2.705×10^{-7}
0.8	0.970	5.736×10^{-7}
1.0	1.000	6.030×10^{-7}

Table 1: Comparison of the Fermi-function difference and current values at identical drain voltages, evaluated at $\mathcal{E} = 0.2\,\mathrm{eV}$.

The ratio of $[f_1(E+U) - f_2(E+U)]_{E=\mathcal{E}}$ at $V_D = 1.0$ to that at $V_D = 0.8$, and the ratio at $V_D = 0.8$ to that at $V_D = 0.5$, are found to be 1.031 and 2.205, respectively.
Similarly, the corresponding current ratios are 1.051 and 2.12. These results confirm that the current scales proportionally with the difference in the Fermi functions.

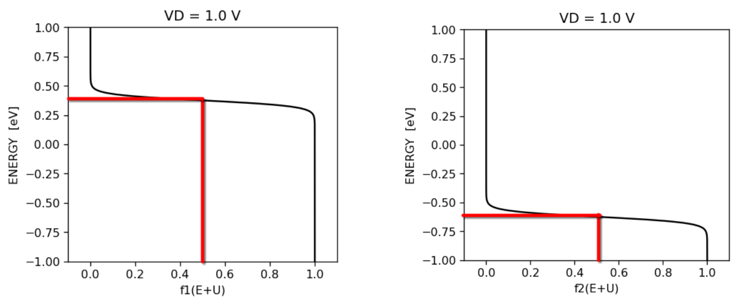


Figure 3: Step locations of the Fermi functions at $V_D = 1.0\,\mathrm{V}$.

At $V_D = 1.0\,\mathrm{V}$, the step locations of the Fermi functions were determined to be $\mu_1 = 0.3760$ and $\mu_2 = 0.6240$. Using these values in `problem1c3.py` yields a self-consistent potential of $U = -0.250\,27\,\mathrm{eV}$.

At this drain voltage, the source contact (f_1) favors electron injection into the channel, while the drain contact (f_2) favors electron absorbtion from the channel. This can be visualized in Figure 3 where the Fermi function of f_1 steps down at a higher energy than f_2. These two biases produce current flow through the channel. Therefore, energies that $\mathcal{E}$ can be located must be less than $\mu_1 = 0.3760$ and greater than $\mu_2 = 0.6240$.

When $V_D = 0$, the electrochemical potentials of the source and drain are equal ($\mu_1 = \mu_2$), which implies that their Fermi functions are identical ($f_1(E) = f_2(E)$). Consequently, the source is trying to fill (or empty) the channel level to the exact same extent that the drain is. The lack of a difference in agenda leads to no net movement of electrons, and thus the current is zero.

2 Current in a Nanoscale MOSFET

The plots seen in Figure 4 were produced by the code provided in `problem2.py`. These figures apply for $V_G = 0.5\,\mathrm{V}$.

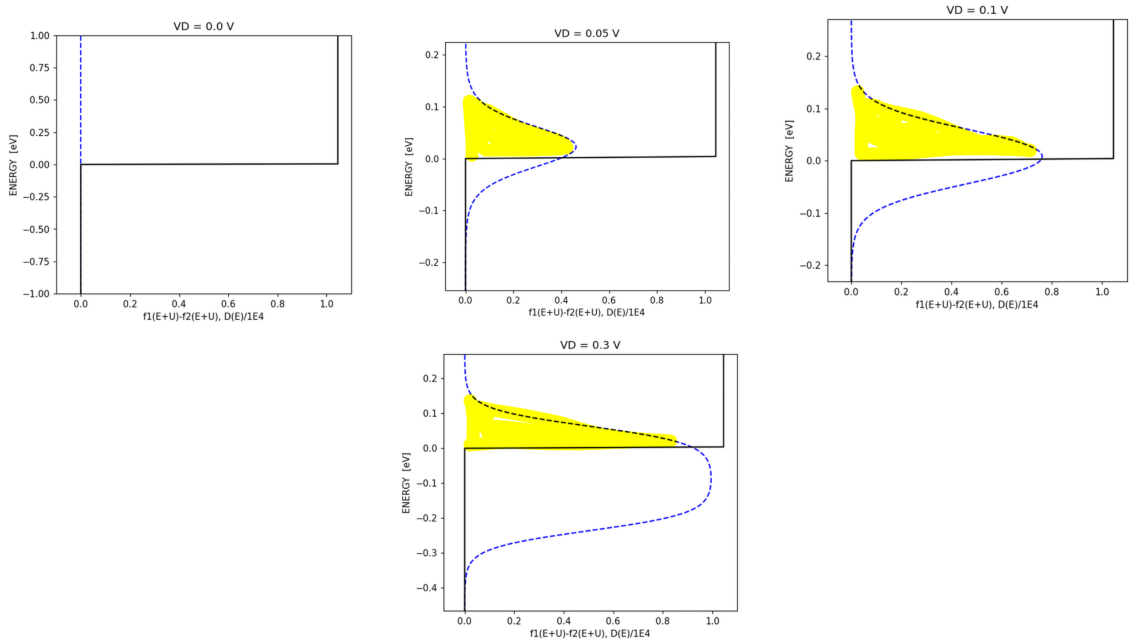


Figure 4: The highlighted regions represent the area under the $f_1(E+U) - f_2(E+U)$ curve that determines current.

As V_D increases, the current saturates because the area under $f_1(E+U) - f_2(E+U)$ remains constant for the region $E \geq 0$. Any further expansion of the curve occurs in the $E < 0$ region, which does not contribute to the total charge transport.

By observing Figure 4 it can be seen that the levels involved in the transport of electrons are in the range $0.2\,\mathrm{eV} \geq E \geq 0.0\,\mathrm{eV}$ based on the bounds of the integral of $f_1(E+U) - f_2(E+U)$ for all energies $E \geq 0$. The energy at which the difference in agenda between the contacts is greatest is $-0.0875\,\mathrm{eV}$, because the peak of $f_1(E+U) - f_2(E+U)$ falls on this point. Looking at the plots of the individual Fermi functions, f_1 settles at its maxima and f_2 settles at its minima at the same point.

To maximize the MOSFET's drain current, material A, who's states vanish at an energy of $-0.4\,\mathrm{eV}$ should be selected. This is because after satisfying $V_{DS} > V_{GS} - V_T$ the area in which the curve intersects the region $E \geq 0$ stays constant for any further increase in V_D. In order to increase the area, hence the current, the region has to be shifted down to a value below 0.

3 Thermoelectric Current

Consider a channel with a Lorentzian density of states $D(E)$ given by:

$$D(E) = D_{\mathcal{E}}(E) = \frac{(\gamma_1 + \gamma_2)/2\pi}{(E - \mathcal{E})^2 + [(\gamma_1 + \gamma_2)/2]^2} \tag{5}$$

where $\gamma = \gamma_1 + \gamma_2 = 0.01\,\text{eV}$. We calculate the current I flowing between the source and drain contacts held at different temperatures, $k_B T_1 = 0.025\,\text{eV}$ and $k_B T_2 = 0.026\,\text{eV}$, with a common equilibrium Fermi level $\mu = 0\,\text{eV}$. The current is given by:

$$I = \frac{q}{\hbar} \frac{\gamma_1 \gamma_2}{\gamma_1 + \gamma_2} [f_1(E) - f_2(E)] \bigg|_{E=\mathcal{E}} \tag{6}$$

Since we are ignoring self-consistency, the potential U is assumed to be zero, and the channel energy level $\mathcal{E}$ is varied from $-0.25\,\text{eV}$ to $0.25\,\text{eV}$. `problem3.py` was written to implement this system and produce the following plots

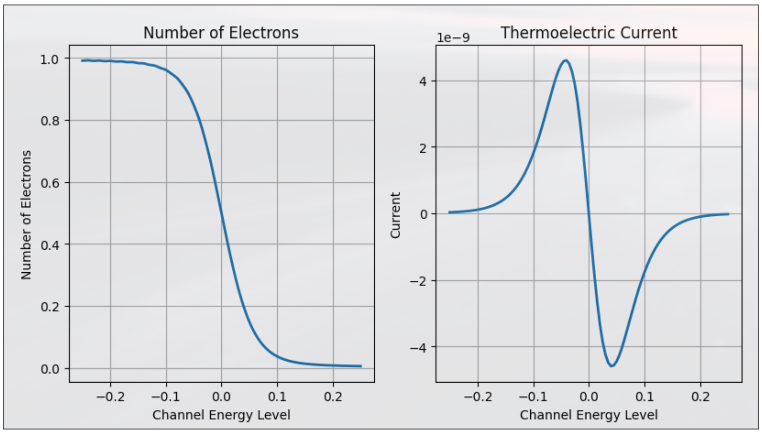


Figure 5: Thermoelectric current (I) and number of electrons (N) as functions of channel energy level ($\mathcal{E}$).

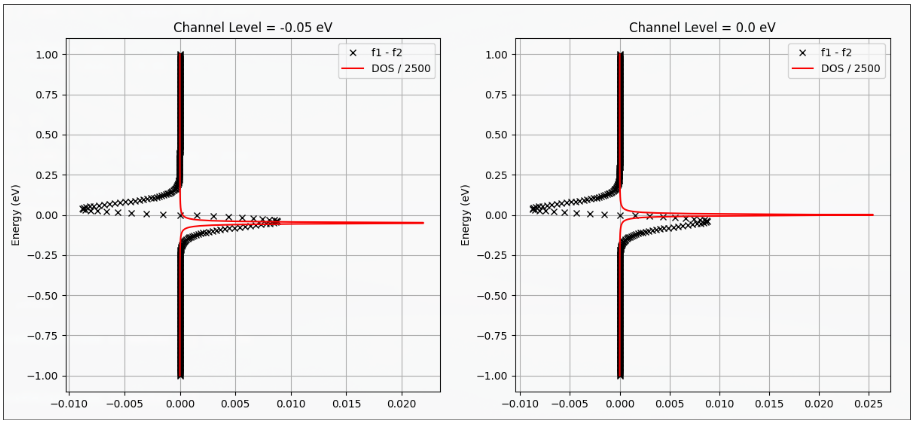


Figure 6: The density of states divided by 2500 plotted with the difference between the Fermi functions of the contacts at $\mathcal{E} = -0.05$eV

The shape of the thermoelectric current in Figure 5 can be attributed to the difference in the steepness of the contact's Fermi functions due to variation in their thermal energy KBT. This causes $f2 - f1$ to adopt a dip/bump shape which carries over into the current due to their proportationality, as shown in previous sections. This is further illustrated by Figure **??** where the variation in thermal energy is exaggerated to show how a change in steepness can produce a positive region below μ and a negative region above μ. Therefore to achieve the maximum value, $\mathcal{E}$ should be positioned below μ.

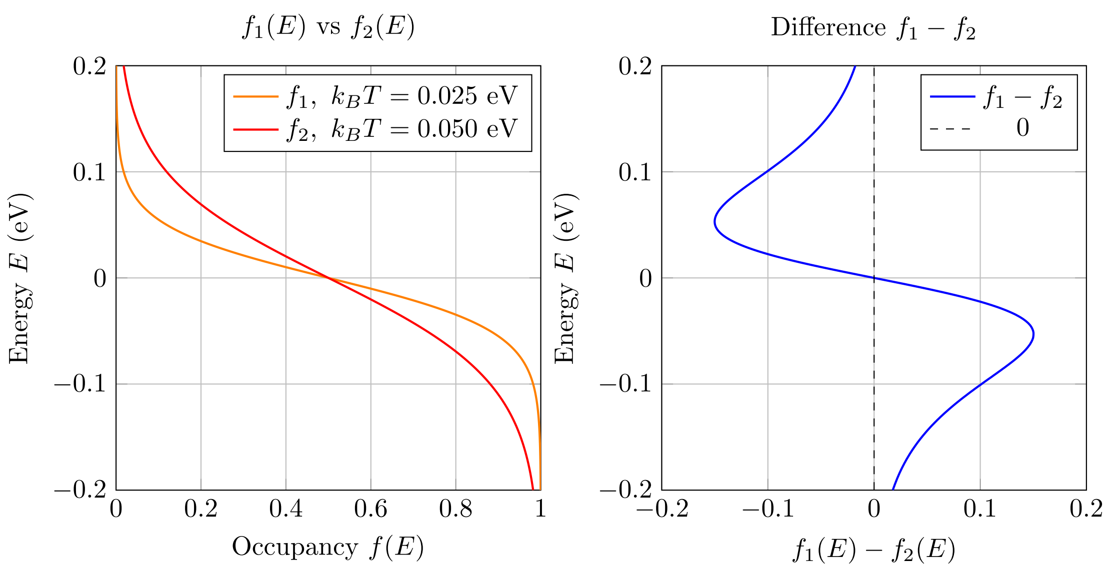


Figure 7: Continuous Fermi–Dirac functions with different thermal energies. The following plots were sketches transformed into LaTex using ChatGPT.

4 Applying What We Learned

4.1 Operation at Absolute Zero

We consider a system operating at absolute zero temperature ($T = 0\,\text{K}$). The channel is characterized by a discrete energy level $\mathcal{E}$, described by the density of states function $D(E) = \delta(E - \mathcal{E})$. The coupling coefficients to the source and drain contacts are symmetric and given by:

$$\gamma_1 = \gamma_2 = 0.005\,\text{eV}$$

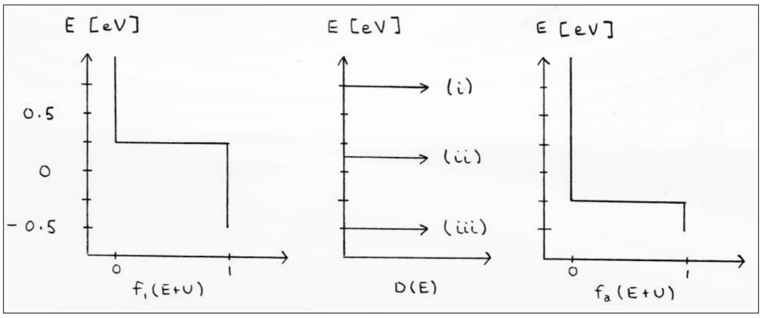


Figure 8: Plots provided for the absolute zero system.

To calculate the current I and the number of channel electrons N for each of these three cases, the difference between f_1 and f_2 along with the density of states $D(E) = \delta(E - \mathcal{E})$, were analyzed in the context of Equation **??** and 4.

For (i) and (iii) it is visibly discernable that $D(E)$ intersects where the Fermi functions are equal, therefore there is no difference in agenda resulting in $I = 0$. However, for (i) the Fermi functions are both at 0 resulting in $N = 0$, while for (ii) both are at 1 resulting in $N = 1$.

At (ii), since $D(E)$ intersects with the $f2 - f1$, leading to a difference of agenda. Because the difference between $f2 - f1 = 1$, current can be calculated using

$$I = \frac{q}{\hbar} \frac{\gamma_1 \gamma_2}{\gamma_1 + \gamma_2} = 0.608\text{nA}$$

Since $\gamma_1 = \gamma_2$, the number of channel electrons (N) is simply

$$N = \frac{f_1(E+U) + f_2(E+U)}{2} = 0.5$$

4.2 Self-Consistent Potential and Voltage Extraction

Given a self-consistent potential $U = -0.25\,\text{eV}$ and an equilibrium Fermi level $\mu = 0$. The source and drain voltages, V_S and V_D, are determined by identifying the electrochemical potentials, μ_1 and μ_2, from the steps in the Fermi functions plotted .

If the graph showed $f_1(E+U)$ and $f_2(E+U)$ as is, using the code in `problem4b.py` the source and drain voltage were found to be $V_S = 0\text{V}$ and $V_D = 0.5\text{V}$.

However if the graph showed f_1 and f_2, using the code in `problem4b.py` the source and drain voltage were found to be $V_S = 0.25\text{V}$ and $V_D = -0.25\text{V}$

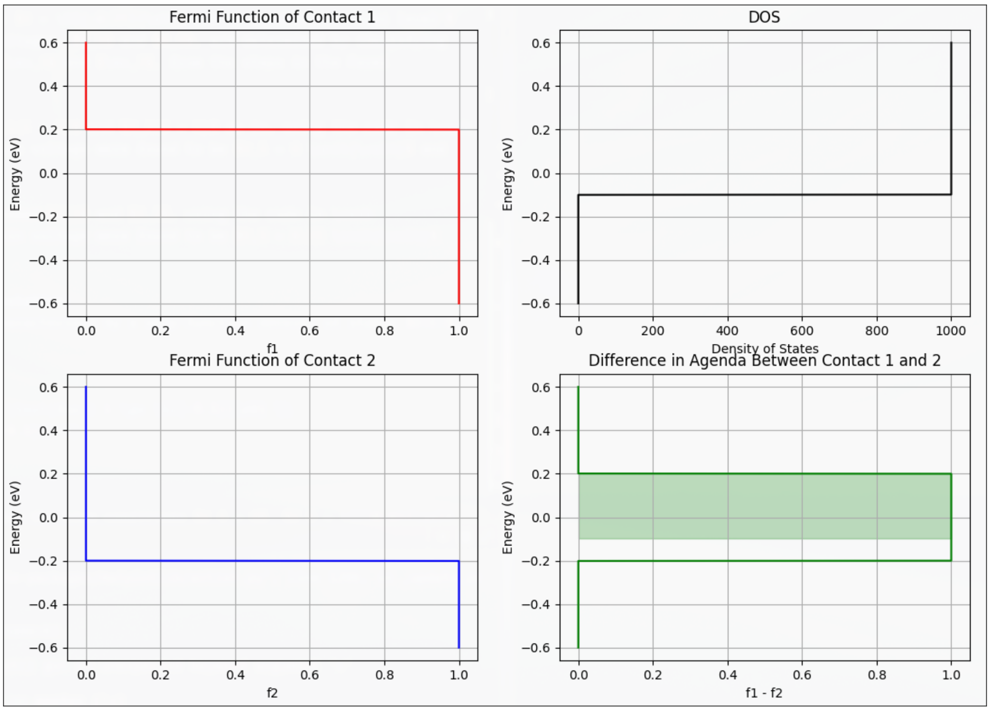
4.3 Nanoscale MOSFET Calculation

We consider a nanoscale MOSFET characterized by a step-function density of states $D(E)$:

$$D(E) = \begin{cases} 10^4 \text{ states/eV} & \text{for } E \geq -0.1\,\text{eV} \\ 0 & \text{for } E < -0.1\,\text{eV} \end{cases}$$

The device operates under the bias conditions $V_S = 0$, $V_G = 0.6\,\text{V}$, and $V_D = 0.4\,\text{V}$. Given these parameters, the self-consistent potential is determined to be $U = -0.2\,\text{eV}$, with the equilibrium Fermi level located at $\mu = 0$.

Assuming operation at $T = 0\text{K}$, `problem4c.py` produced the following plots



i

Figure 9: Plots for the system assuming $T = 0\text{K}$, the shaded area determines the current.

If $\gamma_1 = \gamma_2 = 0.005eV$, to calculate the current, first we determine the energy range of the Fermi window at $T = 0\,\text{K}$. The window extends from the drain potential to the source potential, adjusted by the self-consistent potential U:

$$E_{\text{upper}} = \mu_1 - U = 0 - (-0.2) = 0.2\,\text{eV}$$
$$E_{\text{lower}} = \mu_2 - U = -0.4 - (-0.2) = -0.2\,\text{eV}$$

The density of states $D(E)$ is non-zero (10^4) only for $E \geq -0.1\,\text{eV}$. Therefore, the effective integration range is restricted to $[-0.1\,\text{eV}, 0.2\,\text{eV}]$.
Substituting the given values $\gamma_1 = \gamma_2 = 0.005\,\text{eV}$ into Equation 4:

$$I = \frac{q}{\hbar}\frac{\gamma_1\gamma_2}{\gamma_1+\gamma_2}\int_{-0.1}^{0.2} D(E)\;\mathrm{d}E$$
$$I = \frac{q}{\hbar}\left(\frac{0.005^2}{0.01}\text{eV}\right)\left[10^4 \times (0.2 - (-0.1))\right]$$
$$I = \frac{q}{\hbar}(0.0025\,\text{eV})(3000)$$
$$I = \frac{q}{\hbar}(7.5\,\text{eV})$$

Converting $7.5\,\text{eV}$ to Joules ($1.2015 \times 10^{-18}\,\text{J}$) to solve for the current in Amperes:

$$I = \frac{1.602 \times 10^{-19}}{1.055 \times 10^{-34}} \times 1.2015 \times 10^{-18} \approx 1.82\,\text{mA}$$

At a specified V_G and V_S and a sufficiently high drain voltage, the self-consistent potential saturates to a value of $-0.3\,\text{eV}$. The current is determined by the area under the curve, and since area changed by a factor of $4/3$ the new current can be determined by multiplying the old current by a factor of $4/3$. This produces a value of $I = 2.43\,\text{mA}$.

5 Calculation of Coupling Coefficient γ_2

We consider a nanodevice with a density of states approximated by a delta function $D(E) = \delta(E - \mathcal{E})$, where the energy level is located at $\mathcal{E} = 0.2\,\text{eV}$. The device operates at room temperature ($k_BT = 0.025\,\text{eV}$) with an equilibrium Fermi level $\mu = 0$.
The electrical parameters and bias conditions are given as:

- Biases: $V_G = V_S = 0\,\text{V}$, $V_D = 0.6\,\text{V}$.
- Capacitance parameters: $\alpha_G = \alpha_D = 0.5$, $\alpha_S = 0$.
- Charging energy: $U_0 = 0.25\,\text{eV}$.
- Reference electrons: $N_0 = 0$.

The self-consistent potential U is the sum of the Laplace potential (U_L) and the Poisson potential (U_P), producing the following equation

$$U = U_L + U_P = -(\alpha_G V_G + \alpha_S V_S + \alpha_D V_D) + U_0(N - N_0) \tag{7}$$

Given the self-consistent electron number $N = 0.325$ and the source coupling $\gamma_1 = 5.00 \times 10^{-3}\,\text{eV}$, we rearrange Equation 3 to solve for γ_2

$$\gamma_2 = \gamma_1 \frac{f_1(\mathcal{E}+U) - N}{N - f_2(\mathcal{E}+U)} \tag{8}$$

Using the Equations 7 and 8, first we compute the self-consistent potential U:

$$U = -[0.5(0) + 0(0) + 0.5(0.6)] + 0.25(0.325 - 0)$$
$$U = -0.3 + 0.08125 = -0.218\,75\,\text{eV}$$

Next, we evaluate the Fermi functions at the shifted energy level $E' = \mathcal{E} + U = 0.2 - 0.21875 = -0.018\,75\,\text{eV}$. With $\mu_1 = 0$ and $\mu_2 = -qV_D = -0.6\,\text{eV}$:

$$f_1 = \frac{1}{1 + e^{(E'-\mu_1)/k_BT}} = \frac{1}{1 + e^{-0.01875/0.025}} = \frac{1}{1 + e^{-0.75}} \approx 0.6792$$
$$f_2 = \frac{1}{1 + e^{(E'-\mu_2)/k_BT}} = \frac{1}{1 + e^{(-0.01875+0.6)/0.025}} = \frac{1}{1 + e^{23.25}} \approx 0$$

Finally, substituting these values into Equation 8:

$$\gamma_2 = (5.00 \times 10^{-3})\frac{0.6792 - 0.325}{0.325 - 0} \approx 5.45 \times 10^{-3}\,\text{eV}$$